Manufacturing Quality Control Knowledge Base - 10 Cases

Case ID: C001
Temperature: 250 C
Pressure: 120 bar
Vibration: 5.5 mm/s
Observed Defect: Surface Crack
Root Cause: Excessive Temperature
Corrective Action: Reduce temperature by 10 C.

Case ID: C002
Temperature: 255 C
Pressure: 125 bar
Vibration: 5.8 mm/s
Observed Defect: Surface Crack
Root Cause: High Heater Setting
Corrective Action: Reduce heater setting and increase cooling.

Case ID: C003
Temperature: 195 C
Pressure: 80 bar
Vibration: 2.5 mm/s
Observed Defect: Dimensional Error
Root Cause: Low Pressure
Corrective Action: Increase pressure to 95 bar.

Case ID: C004
Temperature: 200 C
Pressure: 82 bar
Vibration: 2.7 mm/s
Observed Defect: Dimensional Error
Root Cause: Machine Calibration Issue
Corrective Action: Recalibrate pressure controls.

Case ID: C005
Temperature: 240 C
Pressure: 115 bar
Vibration: 5.0 mm/s
Observed Defect: Surface Roughness
Root Cause: Excessive Vibration
Corrective Action: Inspect spindle and bearings.

Case ID: C006
Temperature: 242 C
Pressure: 118 bar
Vibration: 5.2 mm/s
Observed Defect: Surface Roughness
Root Cause: Tool Wear
Corrective Action: Replace worn cutting tool.

Case ID: C007
Temperature: 210 C
Pressure: 95 bar

Vibration: 3.1 mm/s
Observed Defect: Poor Finish
Root Cause: Tool Wear
Corrective Action: Replace tool and verify speed.

Case ID: C008
Temperature: 205 C
Pressure: 92 bar
Vibration: 3.0 mm/s
Observed Defect: Poor Finish
Root Cause: Improper Feed Rate
Corrective Action: Adjust feed rate settings.

Case ID: C009
Temperature: 265 C
Pressure: 135 bar
Vibration: 6.5 mm/s
Observed Defect: Product Rejection
Root Cause: Multiple Parameter Deviations
Corrective Action: Reduce temperature and vibration.

Case ID: C010
Temperature: 270 C
Pressure: 140 bar
Vibration: 7.0 mm/s
Observed Defect: Product Rejection
Root Cause: Process Instability
Corrective Action: Inspect machine and recalibrate.